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Proboscidea Morphology



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Synonyms

[Anatomy](#); [Elephants](#); [Function](#); [Shape](#); [Structure](#)

Definition

General structure of elephants, the only extant members of the order Proboscidea.

being the Sirenians (Sukumar [2003](#)). Proboscidea originated approximately 60 million years ago, during the late Paleocene Epoch; however, only the three extant species remained by the end of the Pleistocene, during which over half of large mammals faced extinction (Sukumar [2003](#)). Proboscideans went through extreme dwarfism and changes of shape before an overall increase in body size around 10,000 years ago (Sukumar [2003](#)). Though Proboscidea fossils have been found on every continent except for Australia and Antarctica (Sukumar [2003](#)), *Loxodonta* and *Elephas* share their origins in Africa during the late Miocene (Todd [2010](#)). It is believed *Elephas* left Africa during the last ice age in the Northern Hemisphere, roughly 35,000 years ago (Spinage [1994](#)).

Introduction

More than 160 Proboscidean species have existed over time (Sukumar [2003](#)); however, only three remain (Payne [2003](#)). The three extant members of the Proboscidea order are as follows: the Asian Elephant (*Elephas maximus*), the African Savannah Elephant (*Loxodonta africana*), and the African Forest Elephant (*Loxodonta cyclotis*) (de Silva [2010](#)). Elephants are the largest terrestrial animals on the planet, and as such they have adapted a unique morphology to support such size. Elephants' closest relatives are sirenians, hyraxes, and armadillos, the closest of these

Skeletal

Elephants are the archetype of the graviportal animals, or animals with large body size and long, columnar limbs (Hutchinson et al. [2006](#)). African elephant cows average a height of 2.7 m tall and a weight of 2.5 tons; while African bulls average a height of just over 3 m tall, and a weight of 5.5 tons (Spinage [1994](#)). Asian elephant cows average a height of 2.35 m tall and a weight of up to 2.54 tons; while Asian bulls average a height of 2.7 m and a weight of up to 4.16 tons (Spinage [1994](#)). Elephants may be able to grow throughout the majority of their lives (Smuts and

Bezuidenhout 1994). The tails of elephants can measure up to 1.5 m in length (Spinage 1994).

The femur is the longest bone in the elephant skeleton, capable of measuring over 1 m long (Smuts and Bezuidenhout 1994). An elephant's skeleton weighs roughly 16.5% of its total living weight (Spinage 1994). According to Gary West, elephant skeletons consist of over 200 bones, though Asian and African elephants differ slightly in structure; Asian elephants typically have 19 rib pairs and 33 caudal vertebra, whereas African elephants have 21 rib pairs and 26 caudal vertebra (Fowler and Mikota 2006). Elephants support about 60% of their weight on their fore-limbs (Spinage 1994). The bones that make up elephant limbs are very dense and have no bone marrow cavity, which make the bones stronger and more able to carry such great weight (Fowler and Mikota 2006).

Brain

Animals with large brain size relative to body size exhibit heightened ability process and apply complex inputs (Garstang 2015). With weights ranging from 3.6 to 6.5 kg, elephants have the heaviest brains of any land mammal (Fowler and Mikota 2006). The heaviest elephant brain ever recorded was 9.0 kg (Garstang 2015). The elephant brain is well-developed, and similarities between human and elephant brains emphasize a capability for higher thinking (Garstang 2015). Within the elephant brain, the cerebellum accounts for 18.6% of the entire brain, a relatively large proportion compared to humans (10.3%); this large cerebellum helps coordinate movements, locomotion, and posture and helps coordinate the complex movements associated with moving the trunk (Garstang 2015).

Elephants have large temporal lobes, associated with memory and learning, and large olfactory lobes, exhibiting a dependence on memory and olfaction (Fowler and Mikota 2006). The large temporal lobe is likely associated with the ability to communicate via infrasonic methods (Garstang 2015). The relative weight of both the

temporal lobes and the cerebral cortex are the greatest of all mammals (Garstang 2015).

Elephants also have an abnormally large hippocampus, responsible for memory, emotion, spatial memory (Bohbot 2015), and for identifying individuals by way of social and chemical memory (Garstang 2015). Elephants' well-developed hippocampal structures allow for advanced processing ability, and the experimental evidence of Polansky et al. suggests elephants can recall the locations of other individuals relative to their own locations (Polansky et al. 2015).

Reproduction

Temporal glands are unique to elephants and are present in all three species and in both males and females (Fowler and Mikota 2006). The female reproductive tract is about 3 m long, making it the longest of all land mammals; the vestibule, or the urogenital canal, accounts for a third of this length at over 1 m (Hildebrandt et al. 2010). The female reproductive tract consists of two ovaries, Fallopian tubes, bicornuate uterus, vagina, and vulva; the most unique features are the 1-m long urogenital canal and the pronounced clitoris (Sukumar 2003). The vagina can measure up to 50 cm in length and is internally marked by longitudinal folds (Hildebrandt et al. 2010). The hymen ruptures during a cow's first delivery, which separates elephants from most mammals, whose hymens rupture during copulation (Hildebrandt et al. 2010). Birth weights range from 100 kg to 120 kg, and cows carry their calves for 22 months (Spinage 1994).

The bull's penis has an S-shaped structure when erect, allowing it to hook into the forward-directed vagina (Sukumar 2003). Bull elephants lack cooling mechanisms for their testes, unlike other mammals (Spinage 1994). Bull testes are located within the abdomen (Sukumar 2003), and near the kidneys (Spinage 1994). Testes can weigh up to three kilograms each (Spinage 1994).

Integument System

Elephant skin color is usually gray, but may become brown or reddish depending on the substrate used for bathing (Fowler and Mikota 2006). Skin can be up to 3.2 cm thick on their backs and as thin as 1.8 mm on their ears (Fowler and Mikota 2006). Elephant skin contains complex polygonal columns, which act as studs in a wall, strengthening certain areas of the skin, particularly the forehead (Fowler and Mikota 2006). The skin on the trunk, breast, legs, ears, and groin are the thinnest areas (Spinage 1994). Elephant skin may be the largest sensory organ of the elephant (Fowler and Mikota 2006). Asian elephants are typically hairier than the African species, and fetal and newborn elephants are hairier than adult elephants (Fowler and Mikota 2006).

Gastrointestinal

The elephant digestive tract begins with the mouth (which includes the trunk), and then continues into the pharynx, esophagus, stomach, small intestine, large intestine, cecum, rectum, and finally, the anus (Greene et al. 2019). Within the mouth, elephant tongues are attached to the floors of their mouths, and though they cannot protrude, they can fold in center; these tongues can weigh as much as 12 kg (Greene et al. 2019). Elephants lack a gallbladder, as do equids (Greene et al. 2019). The gastrointestinal tract of the elephant is approximately three times its body length, which is relatively short when compared to other herbivores; for example, the gastrointestinal tract of the horse is nearly 12 times its body length (Greene et al. 2019).

Similar to rabbits and horses, elephants are hind-gut fermenters, meaning the first significant fermentation site in the digestive tract is the cecum, and the main digestive area is the colon (Fowler and Mikota 2006). This inefficient method of digestion means that elephant must consume large amounts of low-quality forage in short amounts of time; therefore, wild elephants spend between 16 and 20 h eating each day (Fowler and Mikota 2006). The elephant cecum

accounts for approximately 12% of its body weight (Greene et al. 2019). This means that the cecum of 10,000 pound elephant would weigh 1,200 pounds! Asian elephants are capable of digesting between 40% and 50% of their diet, whereas African elephants may digest as little as 22% (Greene et al. 2019). Elephants may defecate between 15 and 12 times daily, with average defecation weight of approximately 10 kg (Fowler and Mikota 2006).

Elephant Trunk Morphology

The elephant trunk has quite the interesting anatomy as it is often perceived as a hand or nose, but on the inside appears as a tongue. One of the most commonly cited elephant facts in literature, media, and elephant outreach is *an elephant trunk has 40,000 muscles, or 150,000 muscle fibers* (Spinage 1997). However, this was only an estimation made by a French Naturalist Georges Cuvier at the beginning of the nineteenth century (Cuvier 1806). There is no current evidence of a proper estimation of the number of muscles inside of a trunk. Elephants have developed trunks to increase bite volume and bite mass. This allows them to increase their food consumption rate (Pretorius et al. 2016). The lack of any bone inside of the elephant trunk makes it very similar to that of octopus arms and tongues; these three examples are known as muscular hydrostats (Kier and Smith 1985).

Three muscle types are found along the trunk including radial muscles located around the nasal passageways, longitudinal muscles located between the radial muscles and skin, and finally oblique muscles located on the ventral side of the trunk below the radial muscles (Kier and Smith 1985). These muscles allow the trunk to elongate, bend, and twist (Kier and Smith 1985). The two oblique muscles allow the trunk to move to the left or right. The most detailed account of elephant trunk muscle types is shown in the original 1909 monograph on the elephant head by JV Boas and Simon Paulli (Boas 1908). Based on cross-sectional observations of elephant trunks, octopi's arms, and various tongues, Kier (Kier and Smith

1985) posited that muscular hydrostats should satisfy constant volume when moving. It has been approximated these muscles contain a modulus of elasticity of 938 kPa through weight lifting experiments (Wilson et al. 1991).

On the outside of the muscle there is a thick layer of elephant skin that appears to be wrinkled. The wrinkled skin on the elephant functions different than that of the skin on the body, which assists with heat regulation and repelling insects (Martins et al. 2018). The skin of the elephant trunk is comprised of wrinkles near the tip that turn into folds as they approach the base of the elephant trunk. Both the skin on the elephant's body and trunk does not contain sweat glands (Spearman 1970). This is one of the reasons the elephant is constantly spraying itself with water and throwing dirt on their body for aiding thermoregulation.

Underneath the wrinkled skin of the elephant trunk are hairs, specifically vibrissal hairs (Spearman 1970). These vibrissal hairs are commonly referred to as whiskers, and they are sensing or tactile hairs (Prescott et al. 2011). It is unknown currently what the use and function of the whiskers along the elephant trunk are; however, there is a large cluster of the hairs near the tip of the elephant trunk.

Their prehensile trunks can pick up objects as small as a peanut (Onondera and Hicks 1999) and manipulate tools (Hart et al. 2008), demonstrating precise motor control. Anatomically, elephants have underdeveloped occipital lobes compared to their olfactory and temporal lobes (Shoshani et al. 2006), which may indicate that they rely less on vision to accomplish motor tasks. Elephants can detect TNT with better accuracy than most trained bomb detection dogs (Miller et al. 2015). In fact, their ability to reach for food items may be impaired if the trunk cannot be used to sense olfactory cues (Foerder et al. 2011). However, elephants likely still utilize vision to accomplish reaching. Their retinas have a concentration of retinal ganglion cells in the upper temporal region, which corresponds to the area of the visual field where the trunk moves (Stone and Halasz 1989). Additionally, they can accurately discriminate

visual cues that are less than a degree in width of the visual field (Shyan-Norwalt et al. 2010).

Ears

African elephant ears are large, positioned above the neck, and shaped similarly to the continent of Africa, while Asian elephant ears are significantly smaller, positioned below the neck, and generally trapezoidal in shape (Fowler and Mikota 2006). Ears are supported by "lattice-like" areas of cartilage and they are used for more than just hearing; they are used for thermoregulation, balance, and communication (Fowler and Mikota 2006). Although they struggle with high-frequency sounds, elephants can hear low-frequency sounds more effectively than any other animal (Fowler and Mikota 2006). Utilizing infrasonic sounds, elephants can communicate over great distances (Fowler and Mikota 2006).

In all three elephant species, ears maintain a high surface area to volume ratio (Wright 1984). Elephant ears are important in thermoregulation because by flapping their ears, African elephants can increase up to 20% surface area for heat dissipation, and Asian elephants can increase their surface area by up to 10% (Fowler and Mikota 2006). Ears have a significant and extensive vascular supply, and are fanned often in warm temperatures (Wright 1984). Arteries in elephant ears constrict under cool weather conditions, and elephants tuck their ears against their sides to maintain body temperature (Wright 1984).

Feet

Elephants are the largest land mammals, and their feet are uniquely fitted to accommodate massive weight (Weissengruber et al. 2006). Movement allows elephants to circulate blood from their feet back to the heart (Fowler and Mikota 2006). The bones of the foot lay on a unique fatty pad, or cushion, that aids in shock absorption and allows elephants to walk without stomping (Spinage 1994). The cushion is composed of sheets and strands of connective tissue that form large and

small compartments filled with adipose tissue (Weissengruber et al. 2006). The cushion spreads when weight is upon it, allowing for the hefty load. It also reduces suction when walking through deep mud because the entire circumference of the foot shrinks as the elephant lifts its foot (Spinage 1994). In addition to weight distribution, the elephant foot's cushion may store or absorb mechanical force (Weissengruber et al. 2006).

The bones of an elephant's foot form an arch, similar to a human foot (Weissengruber et al. 2006). Each foot has five digits, though the amount of phalanges and nails vary by individual elephant (Fowler and Mikota 2006). Asian elephants typically have five nails on the front feet, and four on the rear, while African elephants typically have four on the front feet, and three on the rear (Spinage 1994). The sole of the hindfoot is smaller and more oval than the rounder front foot (Weissengruber et al. 2006).

The foot covering is composed of skin, toenails, and a cornified, though flexible sole called the slipper (Fowler and Mikota 2006). Toenails are large, relatively flat, and are embedded against the foot (Spinage 1994). Similar to humans, each elephant toenail has a cuticle, and is not weight bearing like a hoof (Fowler and Mikota 2006).

Dentition

The enamel loops on the teeth of African Elephants are lozenge-shaped, thus the genus *Loxodonta* was named (Greene et al. 2019). Elephant molars have a unique system of eruption and replacement (Todd 2010). Elephants go through six sets of four molars throughout their lives (Fowler and Mikota 2006). Molars are degraded and lost in sections, and replaced by new teeth which push from behind the old tooth (Fowler and Mikota 2006). Each new tooth is larger than the previous tooth, and each adult elephant molar may weigh greater than five kilograms (Fowler and Mikota 2006). Because of this unique system of replacing teeth, it is possible to approximate an elephant's age by examining its teeth and determining which set the elephant is on (Lindeque 1990).

All African elephants and male Asian elephants have prominent tusks, and female Asian elephants may have small tushes, or vestigial tusks (Fowler and Mikota 2006). Permanent tusks develop within 6–12 months of age, replacing deciduous tusks with no apparent function (Fowler and Mikota 2006). Tusks emerge from a sulcus on each side of the upper lip and are extensions of the upper second incisors (Fowler and Mikota 2006). Male tusks may grow as much as six times faster than female tusks, and all tusks average a growth rate of roughly 17 cm annually (Fowler and Mikota 2006). Elephant and hyrax tusks differ from other mammalian tusks in that their tusks develop from incisors instead of canines (Greene et al. 2019). Elephants have many functional uses for their tusks. They are used to dig for water, to strip bark from trees, to break branches, to carry heavy objects, and are used as weapons during conflicts (Fowler and Mikota 2006). As a person is right-handed or left-handed, elephants tend to be either right-tusked or left-tusked, and the dominant tusk is usually apparent in the degree of wear on the tusk (Fowler and Mikota 2006).

Conclusion

Elephants are the largest of all land animals and as such have developed unique morphologies to support such size and intelligence. From a highly capable brain to heavily adapted trunks and feet, elephants are truly unique.

Cross-References

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